Double-click (or enter) to edit

```
!unzip /content/Pothole.dataset.yolov11.zip
```

```
Archive:  /content/Pothole.dataset.yolov11.zip
replace Pothole.v1-raw.yolov11/data.yaml? [y]es, [n]o, [A]ll, [N]one, [r]ename:
```

```
pip install ultralytics
```

```
Requirement already satisfied: ultralytics in /usr/local/lib/python3.10/dist-packages (8.3.49)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (3.8.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (4.10.0.84)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (11.0.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (6.0.2)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.32.3)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.13.1)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.5.1+cu121)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (0.20.1+cu121)
Requirement already satisfied: tqdm>=4.64.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (4.66.6)
Requirement already satisfied: psutil in /usr/local/lib/python3.10/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.10/dist-packages (from ultralytics) (9.0.0)
Requirement already satisfied: pandas>=1.1.4 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.2.2)
Requirement already satisfied: seaborn>=0.11.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (0.13.2)
Requirement already satisfied: ultralytics-thop>=2.0.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.0.13)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.55.3)
Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (24.2)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4->ultralytics) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4->ultralytics) (2024.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralytics) (3.4.0
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralytics) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralytics) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralytics) (2024.8.30)
Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralytics) (3.16.1)
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralytics) (4.12.2)
Requirement already satisfied: networkx in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralytics) (3.4.2)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralytics) (3.1.4)
Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralytics) (2024.10.0)
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralytics) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.10/dist-packages (from sympy==1.13.1->torch>=1.8.0->ultralytics) (
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.2)
```

```
from ultralytics import YOLO

# Load a COCO-pretrained YOLO11n model
model = YOLO("yolo11n.pt")
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt'...
100%|██████████| 5.35M/5.35M [00:00<00:00, 15.0MB/s]
```

```
import yaml
import os
import random
import cv2
import matplotlib.pyplot as plt
dir_images = "/content/Pothole.v1-raw.yolov11/train/images"
dir_labels = "/content/Pothole.v1-raw.yolov11/train/labels"
image_files = os.listdir(dir_images)

# Choose 16 random image files from the list
random_images = random.sample(image_files, 4)
```

```
final_model = model.train(data="/content/Pothole.v1-raw.yolov11/data.yaml", epochs=30, imgsz=640, batch=16, lr0=0.00001, dropout=0.15)
```

```
Ultralytics 8.3.49 🚀 Python-3.10.12 torch-2.5.1+cu121 CPU (Intel Xeon 2.20GHz)
engine/trainer: task=detect, mode=train, model=yolo11n.pt, data=/content/Pothole.v1-raw.yolov11/data.yaml, epochs=30, time=None, patience=1
Overriding model.yaml nc=80 with nc=1

                   from  n    params  module                                        arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv              [3, 16, 3, 2]
```

```
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 32, 3, 2]
  2                  -1  1      6640  ultralytics.nn.modules.block.C3k2            [32, 64, 1, False, 0.25]
  3                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
  4                  -1  1     26080  ultralytics.nn.modules.block.C3k2            [64, 128, 1, False, 0.25]
  5                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
  6                  -1  1     87040  ultralytics.nn.modules.block.C3k2            [128, 128, 1, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256, 3, 2]
  8                  -1  1    346112  ultralytics.nn.modules.block.C3k2            [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 256, 5]
 10                  -1  1    249728  ultralytics.nn.modules.block.C2PSA           [256, 256, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 13                  -1  1    111296  ultralytics.nn.modules.block.C3k2            [384, 128, 1, False]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 16                  -1  1     32096  ultralytics.nn.modules.block.C3k2            [256, 64, 1, False]
 17                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 19                  -1  1     86720  ultralytics.nn.modules.block.C3k2            [192, 128, 1, False]
 20                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 22                  -1  1    378880  ultralytics.nn.modules.block.C3k2            [384, 256, 1, True]
 23        [16, 19, 22]  1    430867  ultralytics.nn.modules.head.Detect           [1, [64, 128, 256]]
YOLO11n summary: 319 layers, 2,590,035 parameters, 2,590,019 gradients, 6.4 GFLOPs

Transferred 448/499 items from pretrained weights
TensorBoard: Start with 'tensorboard --logdir runs/detect/train6', view at http://localhost:6006/
Freezing layer 'model.23.dfl.conv.weight'
train: Scanning /content/Pothole.v1-raw.yolov11/train/labels.cache... 304 images, 161 backgrounds, 0 corrupt: 100%|██████████| 465/465 [00:
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weigh
/usr/local/lib/python3.10/dist-packages/albumentations/__init__.py:24: UserWarning: A new version of Albumentations is available: 1.4.22 (y
  check_for_updates()
val: Scanning /content/Pothole.v1-raw.yolov11/valid/labels.cache... 133 images, 0 backgrounds, 0 corrupt: 100%|██████████| 133/133 [00:00<?
Plotting labels to runs/detect/train6/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=1e-05' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automat
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)
TensorBoard: model graph visualization added ✅
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to runs/detect/train6
Starting training for 30 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/30         0G      1.907      3.453       2.07          2        640: 100%|██████████| 30/30 [07:06<00:00, 14.22s/it]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|██████████| 5/5 [00:34<00:00,  6.84s/it]


      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/30         0G      1.834      2.899      1.975          3        640: 100%|██████████| 30/30 [06:49<00:00, 13.64s/it]
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|██████████| 5/5 [00:36<00:00,  7.32s/it]
```

```
# prompt: write a code to validate the model

from ultralytics import YOLO
import os

# Assuming 'final_model' from the previous code is available
# If not, load the trained model:
final_model = YOLO("/content/runs/detect/train6/weights/best.pt") # Replace with the actual path

# Define the validation dataset path
val_data_yaml = "/content/Pothole.v1-raw.yolov11/data.yaml" # Replace with the actual path to your validation YAML

# Perform validation
results = final_model.val(data=val_data_yaml)

# Print the validation results
results
```

```
Ultralytics 8.3.49 🚀 Python-3.10.12 torch-2.5.1+cu121 CPU (Intel Xeon 2.20GHz)
YOLO11n summary (fused): 238 layers, 2,582,347 parameters, 0 gradients, 6.3 GFLOPs
val: Scanning /content/Pothole.v1-raw.yolov11/valid/labels.cache... 133 images, 0 backgrounds, 0 corrupt: 100%|██████████| 133/133 [00:00<?
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|██████████| 9/9 [00:32<00:00,  3.57s/it]
                   all        133        330      0.345      0.164      0.122     0.0347
Speed: 2.9ms preprocess, 227.4ms inference, 0.0ms loss, 0.6ms postprocess per image
Results saved to runs/detect/val
ultralytics.utils.metrics.DetMetrics object with attributes:

ap_class_index: array([0])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7f0f96215750>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,    0.003003,    0.004004,    0.005005,    0.006006,    0.007007,
0.008008,    0.009009,     0.01001,    0.011011,    0.012012,    0.013013,    0.014014,    0.015015,    0.016016,    0.017017,
0.018018,    0.019019,     0.02002,    0.021021,    0.022022,    0.023023,
```

```
          0.024024,    0.025025,    0.026026,    0.027027,    0.028028,    0.029029,     0.03003,    0.031031,    0.032032,    0.033033,
    0.034034,    0.035035,    0.036036,    0.037037,    0.038038,    0.039039,     0.04004,    0.041041,    0.042042,    0.043043,
    0.044044,    0.045045,    0.046046,    0.047047,
          0.048048,    0.049049,     0.05005,    0.051051,    0.052052,    0.053053,    0.054054,    0.055055,    0.056056,    0.057057,
    0.058058,    0.059059,     0.06006,    0.061061,    0.062062,    0.063063,    0.064064,    0.065065,    0.066066,    0.067067,
    0.068068,    0.069069,     0.07007,    0.071071,
          0.072072,    0.073073,    0.074074,    0.075075,    0.076076,    0.077077,    0.078078,    0.079079,     0.08008,    0.081081,
    0.082082,    0.083083,    0.084084,    0.085085,    0.086086,    0.087087,    0.088088,    0.089089,     0.09009,    0.091091,
    0.092092,    0.093093,    0.094094,    0.095095,
          0.096096,    0.097097,    0.098098,    0.099099,      0.1001,      0.1011,      0.1021,      0.1031,      0.1041,     0.10511,
    0.10611,     0.10711,     0.10811,     0.10911,     0.11011,     0.11111,     0.11211,     0.11311,     0.11411,     0.11512,     0.11612,
    0.11712,     0.11812,     0.11912,
           0.12012,     0.12112,     0.12212,     0.12312,     0.12412,     0.12513,     0.12613,     0.12713,     0.12813,     0.12913,
    0.13013,     0.13113,     0.13213,     0.13313,     0.13413,     0.13514,     0.13614,     0.13714,     0.13814,     0.13914,     0.14014,
    0.14114,     0.14214,     0.14314,
           0.14414,     0.14515,     0.14615,     0.14715,     0.14815,     0.14915,     0.15015,     0.15115,     0.15215,     0.15315,
    0.15415,     0.15516,     0.15616,     0.15716,     0.15816,     0.15916,     0.16016,     0.16116,     0.16216,     0.16316,     0.16416,
    0.16517,     0.16617,     0.16717,
           0.16817,     0.16917,     0.17017,     0.17117,     0.17217,     0.17317,     0.17417,     0.17518,     0.17618,     0.17718,
    0.17818,     0.17918,     0.18018,     0.18118,     0.18218,     0.18318,     0.18418,     0.18519,     0.18619,     0.18719,     0.18819,
    0.18919,     0.19019,     0.19119,
           0.19219,     0.19319,     0.19419,      0.1952,      0.1962,      0.1972,      0.1982,      0.1992,      0.2002,      0.2012,
    0.2022,      0.2032,      0.2042,     0.20521,     0.20621,     0.20721,     0.20821,     0.20921,     0.21021,     0.21121,     0.21221,
    0.21321,     0.21421,     0.21522,
           0.21622,     0.21722,     0.21822,     0.21922,     0.22022,     0.22122,     0.22222,     0.22322,     0.22422,     0.22523,
    0.22623,     0.22723,     0.22823,     0.22923,     0.23023,     0.23123,     0.23223,     0.23323,     0.23423,     0.23524,     0.23624,
    0.23724,     0.23824,     0.23924,
           0.24024,     0.24124,     0.24224,     0.24324,     0.24424,     0.24525,     0.24625,     0.24725,     0.24825,     0.24925,
    0.25025,     0.25125,     0.25225,     0.25325,     0.25425,     0.25526,     0.25626,     0.25726,     0.25826,     0.25926,     0.26026,
    0.26126,     0.26226,     0.26326,
           0.26426,     0.26527,     0.26627,     0.26727,     0.26827,     0.26927,     0.27027,     0.27127,     0.27227,     0.27327,
    0.27427,     0.27528,     0.27628,     0.27728,     0.27828,     0.27928,     0.28028,     0.28128,     0.28228,     0.28328,     0.28428,
    0.28529,     0.28629,     0.28729,
           0.28829,     0.28929,     0.29029,     0.29129,     0.29229,     0.29329,     0.29429,      0.2953,      0.2963,      0.2973,
    0.2983,      0.2993,      0.3003,      0.3013,      0.3023,      0.3033,      0.3043,     0.30531,     0.30631,     0.30731,     0.30831,
    0.30931,     0.31031,     0.31131,
           0.31231,     0.31331,     0.31431,     0.31532,     0.31632,     0.31732,     0.31832,     0.31932,     0.32032,     0.32132,
    0.32232,     0.32332,     0.32432,     0.32533,     0.32633,     0.32733,     0.32833,     0.32933,     0.33033,     0.33133,     0.33233,
    0.33333,     0.33433,     0.33534,
           0.33634,     0.33734,     0.33834,     0.33934,     0.34034,     0.34134,     0.34234,     0.34334,     0.34434,     0.34535,
    0.34635.     0.34735.     0.34835.     0.34935.     0.35035.     0.35135.     0.35235.     0.35335.     0.35435.     0.35536.     0.35636.
```

```
# prompt: write code for giving a img to validate odel

# Assuming 'final_model' from the previous code is available
# If not, load the trained model:
final_model = YOLO("/content/runs/detect/train6/weights/best.pt") # Replace with the actual path

# Define the path to the image you want to validate
image_path = "/content/Pothole.v1-raw.yolov11/train/images" # Replace with the actual path

# Perform inference on the image
results = final_model(image_path)

# Print the results
results

# Visualize the results (optional)
results[0].plot()
plt.show()
```

```
image 1/465 /content/Pothole.v1-raw.yolov11/train/images/img-100_jpg.rf.233751cac3399aa80b5e58a2afbb753a.jpg: 640x640 1 pothole, 266.1ms
image 2/465 /content/Pothole.v1-raw.yolov11/train/images/img-101_jpg.rf.f213d98946c4f8858a5bf3200c89e39c.jpg: 640x640 1 pothole, 215.9ms
image 3/465 /content/Pothole.v1-raw.yolov11/train/images/img-102_jpg.rf.16c56ff8f761aec963d9b57aa1b2d3fd.jpg: 640x640 1 pothole, 231.0ms
image 4/465 /content/Pothole.v1-raw.yolov11/train/images/img-103_jpg.rf.09abf1a8e3477718e143255af3534d29.jpg: 640x640 1 pothole, 215.1ms
image 5/465 /content/Pothole.v1-raw.yolov11/train/images/img-108_jpg.rf.a35e86abc558a98f252bfc10e49fd6d9.jpg: 640x640 2 potholes, 225.8ms
image 6/465 /content/Pothole.v1-raw.yolov11/train/images/img-109_jpg.rf.add373710d0d425bd9340c33878e7d42.jpg: 640x640 1 pothole, 233.1ms
image 7/465 /content/Pothole.v1-raw.yolov11/train/images/img-10_jpg.rf.eaceaaa6f119cf6eb49edc1bc92efaa2.jpg: 640x640 1 pothole, 247.8ms
image 8/465 /content/Pothole.v1-raw.yolov11/train/images/img-110_jpg.rf.0390f81a2c5b2ad5676a8eb5b043d743.jpg: 640x640 1 pothole, 220.0ms
image 9/465 /content/Pothole.v1-raw.yolov11/train/images/img-111_jpg.rf.d7e58630e249c45d8c1d564d847dc236.jpg: 640x640 1 pothole, 251.8ms
image 10/465 /content/Pothole.v1-raw.yolov11/train/images/img-113_jpg.rf.31b41591a5e0c7b0aa02e133e79e1e2a.jpg: 640x640 1 pothole, 220.1ms
image 11/465 /content/Pothole.v1-raw.yolov11/train/images/img-114_jpg.rf.b58901654a45a559a41ffaa25f1bf741.jpg: 640x640 1 pothole, 224.8ms
image 12/465 /content/Pothole.v1-raw.yolov11/train/images/img-115_jpg.rf.ebc1ec2d7bb5caf80d06511d74f9e2da.jpg: 640x640 1 pothole, 225.5ms
image 13/465 /content/Pothole.v1-raw.yolov11/train/images/img-117_jpg.rf.56a7e3b29e60b28e25b3f4a3c7817605.jpg: 640x640 (no detections), 224
image 14/465 /content/Pothole.v1-raw.yolov11/train/images/img-119_jpg.rf.5df904c99a0e765190d9421a4bf04bc6.jpg: 640x640 2 potholes, 226.5ms
image 15/465 /content/Pothole.v1-raw.yolov11/train/images/img-11_jpg.rf.b7a2e18571a53d7512f101f224b7c794.jpg: 640x640 1 pothole, 233.4ms
image 16/465 /content/Pothole.v1-raw.yolov11/train/images/img-121_jpg.rf.d202af2960b0bb1928539a6fcb3cdbfd.jpg: 640x640 2 potholes, 226.1ms
image 17/465 /content/Pothole.v1-raw.yolov11/train/images/img-123_jpg.rf.8ae4c2726246a3dea5d41f6ba3a64639.jpg: 640x640 1 pothole, 231.7ms
image 18/465 /content/Pothole.v1-raw.yolov11/train/images/img-124_jpg.rf.fdc89fc9a29a16b57ad54dd3c63e7952.jpg: 640x640 1 pothole, 222.8ms
image 19/465 /content/Pothole.v1-raw.yolov11/train/images/img-126_jpg.rf.eab9b684c8570533e5cd43517776eeec.jpg: 640x640 2 potholes, 243.4ms
image 20/465 /content/Pothole.v1-raw.yolov11/train/images/img-127_jpg.rf.cda9134fdf2c217082118b38d27fd7ee.jpg: 640x640 1 pothole, 224.5ms
image 21/465 /content/Pothole.v1-raw.yolov11/train/images/img-128_jpg.rf.6d1435fcdbcb19d4ff22f1d865026470.jpg: 640x640 1 pothole, 222.3ms
image 22/465 /content/Pothole.v1-raw.yolov11/train/images/img-129_jpg.rf.fbd5c4841019062274b4e9d1daeab4c3.jpg: 640x640 1 pothole, 215.2ms
image 23/465 /content/Pothole.v1-raw.yolov11/train/images/img-12_jpg.rf.473a95f653f8c7227e4bc5c73f8ed19b.jpg: 640x640 2 potholes, 220.8ms
```

```
image 24/465 /content/Pothole.v1-raw.yolov11/train/images/img-131_jpg.rf.6bc87513d54494e92b56968b81d290b5.jpg: 640x640 1 pothole, 322.2ms
image 25/465 /content/Pothole.v1-raw.yolov11/train/images/img-134_jpg.rf.833f57b2c592a23189208dcb656eaaa2.jpg: 640x640 1 pothole, 332.9ms
image 26/465 /content/Pothole.v1-raw.yolov11/train/images/img-135_jpg.rf.5f0fb06ed79bee89c8378fc74adf59ec.jpg: 640x640 3 potholes, 317.6ms
image 27/465 /content/Pothole.v1-raw.yolov11/train/images/img-136_jpg.rf.3372e97742e4ded8b208f90043981012.jpg: 640x640 (no detections), 331
image 28/465 /content/Pothole.v1-raw.yolov11/train/images/img-137_jpg.rf.5c3cf778bb884760e276baacc8372f66.jpg: 640x640 2 potholes, 320.9ms
image 29/465 /content/Pothole.v1-raw.yolov11/train/images/img-138_jpg.rf.ec6549719f138be69aa49f6cdc068227.jpg: 640x640 1 pothole, 332.2ms
image 30/465 /content/Pothole.v1-raw.yolov11/train/images/img-139_jpg.rf.af2a2cdcef3c3e01d9ad30e9f058e93e.jpg: 640x640 1 pothole, 348.9ms
image 31/465 /content/Pothole.v1-raw.yolov11/train/images/img-13_jpg.rf.a4db51428d0461f9ed82e12eb6d87a03.jpg: 640x640 1 pothole, 324.4ms
image 32/465 /content/Pothole.v1-raw.yolov11/train/images/img-141_jpg.rf.9fb75a7f4520090a669516f6c16c105e.jpg: 640x640 2 potholes, 349.6ms
image 33/465 /content/Pothole.v1-raw.yolov11/train/images/img-142_jpg.rf.324ee78ee3122e5a751d32198a2e11cb.jpg: 640x640 1 pothole, 361.4ms
image 34/465 /content/Pothole.v1-raw.yolov11/train/images/img-143_jpg.rf.b4553a89ca02d904d7d3d8b601d92033.jpg: 640x640 1 pothole, 347.2ms
image 35/465 /content/Pothole.v1-raw.yolov11/train/images/img-144_jpg.rf.2e98a93812eee1ad65ad7e3f9f5620ad.jpg: 640x640 1 pothole, 327.5ms
image 36/465 /content/Pothole.v1-raw.yolov11/train/images/img-148_jpg.rf.eddf84184c0a99958850113c36e7b754.jpg: 640x640 1 pothole, 307.2ms
image 37/465 /content/Pothole.v1-raw.yolov11/train/images/img-149_jpg.rf.44eff0efcbd1ccfa08da8768642f71c9.jpg: 640x640 1 pothole, 227.2ms
image 38/465 /content/Pothole.v1-raw.yolov11/train/images/img-150_jpg.rf.f3ebd01cb708363657a8f21645c06c39.jpg: 640x640 1 pothole, 220.8ms
image 39/465 /content/Pothole.v1-raw.yolov11/train/images/img-151_jpg.rf.52e5bb5d308649772ceef0bfcde24f96.jpg: 640x640 2 potholes, 232.5ms
image 40/465 /content/Pothole.v1-raw.yolov11/train/images/img-152_jpg.rf.621b5126c90ae5176bc3d2415fda4fe8.jpg: 640x640 1 pothole, 250.7ms
image 41/465 /content/Pothole.v1-raw.yolov11/train/images/img-155_jpg.rf.7c1dd28f3db86d175cca8437cece1d08.jpg: 640x640 1 pothole, 238.4ms
image 42/465 /content/Pothole.v1-raw.yolov11/train/images/img-156_jpg.rf.ff820d5c1ff5fc1a1fc27fa5647cc146.jpg: 640x640 1 pothole, 239.6ms
image 43/465 /content/Pothole.v1-raw.yolov11/train/images/img-157_jpg.rf.764e348f9144fed2fd906c23ff15afe5.jpg: 640x640 (no detections), 253
image 44/465 /content/Pothole.v1-raw.yolov11/train/images/img-158_jpg.rf.80110529bda206547f2c5482abfca272.jpg: 640x640 2 potholes, 284.2ms
image 45/465 /content/Pothole.v1-raw.yolov11/train/images/img-159_jpg.rf.9a2234e70393ae42ad518c7c34b031c5.jpg: 640x640 1 pothole, 233.0ms
image 46/465 /content/Pothole.v1-raw.yolov11/train/images/img-15_jpg.rf.b937fa68f73a1ee37d832a279f432941.jpg: 640x640 1 pothole, 216.5ms
image 47/465 /content/Pothole.v1-raw.yolov11/train/images/img-160_jpg.rf.ef4e05f9ef187938ab9bd0321c4dcf4d.jpg: 640x640 1 pothole, 219.7ms
image 48/465 /content/Pothole.v1-raw.yolov11/train/images/img-162_jpg.rf.0459e44683fd65132e9e2558f7cfcc6a.jpg: 640x640 1 pothole, 232.7ms
image 49/465 /content/Pothole.v1-raw.yolov11/train/images/img-163_jpg.rf.232bf43b16b71f5c5581f7d908e3e253.jpg: 640x640 (no detections), 221
image 50/465 /content/Pothole.v1-raw.yolov11/train/images/img-164_jpg.rf.cb6665fd71f9ea53c23dc545be31d2e1.jpg: 640x640 1 pothole, 216.7ms
image 51/465 /content/Pothole.v1-raw.yolov11/train/images/img-165_jpg.rf.a4b4be5d004a62f1bbf126232edb2785.jpg: 640x640 1 pothole, 214.6ms
image 52/465 /content/Pothole.v1-raw.yolov11/train/images/img-167_jpg.rf.a2c9b6a4a769135a8f5f7e671b636ac8.jpg: 640x640 1 pothole, 231.9ms
image 53/465 /content/Pothole.v1-raw.yolov11/train/images/img-169_jpg.rf.9ad23ea6b668a59c48e362859efe3515.jpg: 640x640 2 potholes, 218.2ms
image 54/465 /content/Pothole.v1-raw.yolov11/train/images/img-16_jpg.rf.3018a4883623e53e929435fc084c2200.jpg: 640x640 1 pothole, 224.3ms
image 55/465 /content/Pothole.v1-raw.yolov11/train/images/img-170_jpg.rf.448c70ae76619d82c629389410701f9c.jpg: 640x640 3 potholes, 218.6ms
image 56/465 /content/Pothole.v1-raw.yolov11/train/images/img-172_jpg.rf.d849de955bfb8608552f1adf347f3ef1.jpg: 640x640 1 pothole, 229.7ms
```

```
final_model.export(format="onnx")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-3-864539b0ab04> in <cell line: 1>()
----> 1 final_model.export(format="onnx")

NameError: name 'final_model' is not defined
```

```
# Assuming 'final_model' from the previous code is available
# If not, load the trained model:
from ultralytics import YOLO
final_model = YOLO("/content/runs/detect/train6/weights/best.pt") # Replace with the actual path

#Export the model
final_model.export(format="onnx")
```

```
Ultralytics 8.3.49 🚀 Python-3.10.12 torch-2.5.1+cu121 CPU (Intel Xeon 2.20GHz)
YOLO11n summary (fused): 238 layers, 2,582,347 parameters, 0 gradients, 6.3 GFLOPs

PyTorch: starting from '/content/runs/detect/train6/weights/best.pt' with input shape (1, 3, 640, 640) BCHW and output shape(s) (1, 5, 8400) (

ONNX: starting export with onnx 1.17.0 opset 19...
ONNX: slimming with onnxslim 0.1.44...
ONNX: export success ✅ 2.4s, saved as '/content/runs/detect/train6/weights/best.onnx' (10.1 MB)

Export complete (3.3s)
Results saved to /content/runs/detect/train6/weights
Predict:         yolo predict task=detect model=/content/runs/detect/train6/weights/best.onnx imgsz=640
Validate:        yolo val task=detect model=/content/runs/detect/train6/weights/best.onnx imgsz=640 data=/content/Pothole.v1-raw.yolov11/data.
Visualize:       https://netron.app
```

```
# prompt: write code for download weights of dataset

from google.colab import files
files.download("/content/runs/detect/train6/weights/best.pt")
```